BipedNav: monocular stair and obstacle perception, and virtual-holonomic-constraint walking for two- and five-link bipeds

re-implementation, verification and extension of the 2024 perception poster and the ECE1658 controllers

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September 5, 2026

Abstract

BipedNav couples two pieces of earlier work: a perception layer that recovers staircase geometry (rise h, run b) from a single RGB image through a monocular metric-depth network and plane extraction (Qiu, Narayanan, Cheng, Laschowski, 2024), and a locomotion layer that walks a planar two-link "acrobot" biped with one hip actuator by enforcing a virtual holonomic constraint (VHC) $q_2 = \phi(q_1)$ through feedback linearisation (ECE1658 course project, 2024, originally in MATLAB). Both layers were re-implemented in Python from the archived code and data, tested, and extended: the perception pipeline now identifies treads and risers from surface normals and clusters plane offsets, reaching a mean absolute rise error of 1.7 cm (RealSense cloud) and 2.1 cm (monocular cloud) against tape measurements on the five archived scenes; the walker is simulated as a hybrid system on real terrain with collision checking of the swing foot against treads and risers. The flat-ground design of the report is reproduced to 4×10^{-5} in the polynomial coefficients and its measured return-map slope (0.9329) matches the analytic Floquet multiplier $\delta^2/M^- = 0.9326$. Three findings are new. (i) The impact map of the course template relabels velocities by a reflection $R = -I$, which is not the Jacobian of the leg swap; with the kinematically exact relabelling the same gait retains $\delta = 0.115$ instead of 0.336 of its phase velocity and is no longer a stable limit cycle, although the optimiser still finds exact-map gaits with swing-foot retraction. (ii) A point-foot, equal-leg, knee-less biped can only pass its legs with a vertical stance leg without dipping into a flat tread, which requires an aperture $\beta > 2\alpha$ for stair pitch α and confines the landing point to within $b - \sqrt{2h - h^2}$ (leg units) of the nosing; human-scale stairs ($\alpha \approx 23^\circ$) are out of reach for this morphology, and the report's stair proof of concept relied on the slope model that hides a 5 cm dip at the rear nosing. (iii) Within the reachable regime, the same VHC gaits designed for a slope walk real collision-checked staircases when the foot placement obeys a simple rule: land within 5 % of the tread depth from the nosing when ascending (5° pitch), stand on the nosing when descending (5° and 10°). The locomotion layer was then rebuilt on the five-link biped with knees and torso of Assignment 1 (Section 4): its closed-form dynamics, impact map and leg-swap relabelling are verified to machine precision, its flat-ground gait is a hybrid limit cycle whose measured return-map slope equals the analytic multiplier (0.790), and, from one view of a box obstacle, a planner schedules the approach steps and two step-over gaits chained through the impact map so that the walker clears an 8 cm box end to end in collision-checked simulation; taller boxes and stair climbing expose the limits of the present gait optimiser and are reported as such.

Keywords: monocular depth estimation, stair parameter extraction, virtual holonomic constraints, hybrid zero dynamics, acrobot biped, impact map, stair climbing.

1 Overview

The archived project (`BipedNav/`) contains the perception workspace (RealSense ROS driver, bag processing scripts, five sample scenes with RGB, RealSense depth, monocular depth/normal estimates and hand measurements) and the control code (MATLAB `control/MATLAB/PART1`, a partial Python port). Nothing in the archive could be executed as is: the MATLAB scripts need the Symbolic Math Toolbox session state, the Python port was unfinished, and the perception scripts assume a ROS Noetic install. This document records the re-implementation in `experiments/`:

- `stair_perception.py`: staircase geometry from organised point clouds (Section 2.1);
- `acrobot_vhc.py`: two-link model, impact map, VHC design (linear and optimised), constrained dynamics, terrain model, hybrid simulation (Section 2.2);
- `biped5_vhc.py`, `plan_obstacle.py`, `obstacle_perception.py`: the five-link biped, its Bézier VHC design and gait chaining, the perceive–plan–walk planner and the obstacle extractor (Section 4);
- `design_all.py`, `sweep_stairs.py`: gait design batches; `run_experiments.py`, `fig_perception.py`: every figure and number in this document; `make_video.py`: the demonstration video; `render.py`: plotting in the website's visual language.

The web page `https://thejerrycheng.github.io/bipednav.html` presents the same material.

2 Method

2.1 Perception: stair geometry from one image

The 2024 poster fine-tuned Metric3D v2 [1] on 16 780 RGB-D stair images, back-projected the metric depth with the camera intrinsics and extracted planes by iterative RANSAC. The archived sample set contains, per scene, the RealSense D435 depth (640×480 px, mm), the organised point clouds from that depth ("RealSense cloud", binary PLY) and from the network's depth ("monocular cloud", ASCII PLY), the network's normal map ($3 \times 480 \times 640$), and two tape measurements of rise and run per staircase (`measurements.csv`). The estimated-depth tensors are empty files in the archive; the monocular clouds are used instead.

Iterative RANSAC on these clouds fragments the grazing treads and fits planes across tread/riser edges (the first re-run gave 6–9 cm "rises" on three scenes), so the pipeline was rebuilt around the poster's own idea of predominant normals:

1. *Normals.* Central differences on the image grid (baseline 3 px, jumps > 6 cm rejected), oriented toward the camera; every second pixel is kept ($\approx 7 \times 10^4$ points).
2. *Predominant normals.* A $4°$ spherical histogram of the normals; its modes are refined as means over $18°$ cones. The tread normal $\mathbf{u}$ is the most populated mode pointing up (negative camera-y). The walking direction $\mathbf{f}$ is the optical axis projected on the tread plane, refined by the riser mode if a mode exists $\geq 70°$ from $\mathbf{u}$ and within $30°$ of $-\mathbf{f}$ (this rejects side walls, whose normals are lateral).
3. *Refinement.* $\mathbf{u}$ is re-estimated as the least-squares normal of the most populated tread cluster with a slab shrinking from 4 cm to 1.2 cm.
4. *Offset clustering.* Tread points ($\angle(\mathbf{n}, \mathbf{u}) < 15°$) are projected on $\mathbf{u}$, riser points ($\angle(\mathbf{n}, -\mathbf{f}) < 20°$) on $\mathbf{f}$; a 5 mm histogram smoothed by a Gaussian ($\sigma = 1$ cm) gives the plane offsets as peaks at least 7 cm apart and above 15 % of the maximum.
5. *Lattice fit.* Rise h is the least-squares lattice spacing of the tread offsets (adjacent gaps between 6 and 45 cm); run b is the lattice spacing of the riser offsets (15–60 cm) or, when fewer than two risers are visible, of the front edges (5th percentile along $\mathbf{f}$) of consecutive

treads.

Camera pitch follows from $\mathbf{u}$: $\tan\psi = -u_z/u_y$. The whole pipeline is deterministic and runs in 0.09 s (RealSense) / 0.32 s (ASCII read dominated) per scene on a laptop CPU.

2.2 Acrobot model and impact map

Two identical legs of length $l = 1$ m, mass $m = 1$ kg at $l_c = l/2$, inertia $I_z = ml^2/12$, one torque τ at the hip. q_1 is the absolute angle of the stance leg (foot to hip) from the $+x$ axis and q_2 the relative angle of the swing leg; $B = [0\ 1]^\top$, $B^\perp = [1\ 0]$. The pinned dynamics $D(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) = B\tau$ are closed-form,

$$D = \begin{bmatrix} ml_c^2 + m(l^2 + l_c^2 + 2ll_c\cos q_2) + 2I_z & m(l_c^2 + ll_c\cos q_2) + I_z \\ \cdot & ml_c^2 + I_z \end{bmatrix}, \tag{1}$$

$$C = \begin{bmatrix} \eta\dot{q}_2 & \eta(\dot{q}_1 + \dot{q}_2) \\ -\eta\dot{q}_1 & 0 \end{bmatrix}, \qquad \eta = -mll_c\sin q_2, \tag{2}$$

$G = mg\,[\,(l+l_c)\cos q_1 + l_c\cos(q_1+q_2),\ \ l_c\cos(q_1+q_2)\,]^\top$, and were verified against the sympy derivation of the MATLAB template to 9×10^{-16} (`Model.check_closed_forms`). The impact is the rigid, plastic impact of Westervelt et al. [3]: with the unpinned mass matrix $\bar{D}(q)$ of coordinates (q,x) and $E = [\partial p_{\mathrm{sw}}/\partial q\ \ I_2]$,

$$\begin{bmatrix} \bar{D} & -E^\top \\ E & 0 \end{bmatrix} \begin{bmatrix} \dot{q}^+ \\ F \end{bmatrix} = \begin{bmatrix} \bar{D}\,[I_2\ 0]^\top\dot{q}^- \\ 0 \end{bmatrix}, \qquad \dot{q}_{\mathrm{post}} = \Delta_{\dot{q}}(q^-)\,\dot{q}^-, \tag{3}$$

which was checked to 10^{-15} against conservation of angular momentum about the new stance foot and of the old stance leg about the hip. The legs are then relabelled. The exact swap is $q_1^+ = q_1 + q_2 - \pi$, $q_2^+ = 2\pi - q_2$, whose Jacobian is $R_{\mathrm{kin}} = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$. The course template instead uses $T(q) = -q + [\pi + 2\alpha,\ 2\pi]^\top$ and $\dot{q}^+ = -\dot{q}_{\mathrm{post}}$, i.e. $R_{\mathrm{refl}} = -I$. On the impact surface $q_2 + 2q_1 = 2\pi + 2\alpha$ both give the same configuration, but the two velocity maps agree only for velocities tangent to that surface ($\dot{q}_2 = -2\dot{q}_1$), which post-impact velocities are not (the former stance foot lifts, so $J_{\mathrm{sw}}\dot{q}^+ = -\dot{x}^+ \neq 0$). Physically the new stance leg keeps the angular velocity the old swing leg had, $\dot{q}_1^+ = \dot{q}_1 + \dot{q}_2$; the reflection assigns it $-\dot{q}_1$ instead. Both conventions are implemented (`Model(relabel=...)`) and every result states which one it uses; the impact matrix of the handout is $\mathcal{I} = R\,\Delta_{\dot{q}}(q^-)$.

2.3 VHC design

Following the ECE1658 handout, $q = \sigma(\theta) = [\,q_1^+ - \theta\tilde{q}_1,\ \phi(\theta)\,]^\top$ with $\theta \in [0,1]$, $\tilde{q}_1 = \beta$, $q^\pm = [\,(\pi \pm \beta)/2 + \alpha,\ \pi \mp \beta\,]^\top$ for aperture β and (design) slope α, and ϕ a polynomial of degree $k-1$ satisfying

$$\phi(0) = q_2^+, \quad \phi(1) = q_2^-, \quad \phi(\theta_f) = \pi, \quad \phi'(0) = f(v_1), \quad \phi'(1) = v_1, \quad \phi'(\theta_f) = v_2, \tag{4}$$

where $f(v_1) = -\tilde{q}_1(-\mathcal{I}_{21}\tilde{q}_1 + \mathcal{I}_{22}v_1)/(-\mathcal{I}_{11}\tilde{q}_1 + \mathcal{I}_{12}v_1)$ makes $\sigma'(0) \parallel \mathcal{I}\sigma'(1)$ (hybrid invariance). The report uses $\theta_f = 1/2$; Section 2.6 explains why treads need $\theta_f = 1/2 + \alpha/\beta$. For $k = 6$ the six conditions determine ϕ (a 6×6 Vandermonde solve); for $k > 6$ and for slopes and stairs the design is the nonlinear program of report Part 2/4, solved here with SLSQP after eliminating the equalities analytically: the decision vector is $(v_1, v_2, c_6, \ldots, c_{k-1})$ and the six lowest coefficients follow from (4). Inequalities: transversality margin $\min_\theta(B^\perp D\sigma')^2 \geq \zeta$ without sign change; existence $\delta^2/M^- \leq$

0.95; $V^- < 0$ and the energy margin $-V^-\delta^2/(M^- - \delta^2) \geq 1.05\, V_{\max}$; for flat ground the no-scuff tangent bounds $v_1 < 2\tilde{q}_1$, $f(v_1) < 2\tilde{q}_1$, $v_2 > 2\tilde{q}_1$; and, sampled at 241 phases, positive clearance of the swing foot above the terrain (treads, risers or slope line) except at the three legitimate contacts. The objective is the curve length $\int \|\sigma'\| d\theta$ with a 10^{-5} ridge on the coefficients. A Nelder–Mead penalty polish runs when SLSQP stops infeasible.

2.4 Constrained dynamics, stability test and controller

With $\Psi_1 = -B^\perp G/(B^\perp D\sigma')$ and $\Psi_2 = -B^\perp(D\sigma'' + C\sigma')/(B^\perp D\sigma')$ evaluated on σ, the zero dynamics is $\ddot{\theta} = \Psi_1 + \Psi_2\dot{\theta}^2$, with virtual mass $M(\theta) = \exp(-2\int_0^\theta \Psi_2)$ and potential $V(\theta) = -\int_0^\theta \Psi_1 M$ (cumulative trapezoid on 801 points). With $\delta = \langle\sigma'(0), \mathcal{I}\sigma'(1)\rangle/\|\sigma'(0)\|^2$ the restricted Poincaré map in $\zeta = \frac{1}{2}M^-(\dot{\theta}^-)^2$ is affine, $\zeta_{k+1} = (\delta^2/M^-)\,\zeta_k - V^-$, so the limit cycle exists and is exponentially stable iff $0 < \delta^2/M^- < 1$, $V^- < 0$ and $-V^-\delta^2/(M^- - \delta^2) > V_{\max}$, with $\dot{\theta}^{+*} = \delta\sqrt{-2V^-/(M^- - \delta^2)}$ and Floquet multiplier δ^2/M^-. The controller is the report's input–output linearisation of $y = q_2 - \phi(\theta(q_1))$,

$$\tau = \big(HD^{-1}B\big)^{-1}\Big(HD^{-1}(C\dot{q}+G) - K_p \sin y - K_d H\dot{q} - \dot{H}\dot{q}\Big), \quad H = [\phi'/\tilde{q}_1,\ 1],\ K_p = 40^2,\ K_d = 80. \tag{5}$$

2.5 Terrain and collision model

A staircase is the set of treads $n \in \mathbb{Z}$ at height nh spanning $x \in [d + (n-1)b,\ d + nb)$ relative to the stance foot, so d is the distance from the stance foot to the next riser and $h < 0$ descends; $h = 0$ is flat ground. Its design slope is $\alpha = \arctan(h/b)$ and its aperture $\beta = 2\arcsin(\sqrt{b^2 + h^2}/2l)$; a slope is the line $y = x\tan\alpha$ with stride s. One step is integrated in two phases (DOP853 via `solve_ivp`, rtol 10^{-9}): until the swing foot passes $x = d$ (checking on the dense output that it never goes below the floor except within 5 % of l of the stance foot, where the legs legitimately fold), then until it touches the landing surface; a foot below the next tread when it crosses the riser plane is a *trip*, a floor violation deeper than 1 mm a *scuff*, a hip lower than $0.25\, l$ a *fall*. Then (3) and the relabelling are applied at the actual touchdown configuration.

2.6 Why stairs are hard for a knee-less point-foot biped

Two facts of pure geometry. (a) The swing leg passes the stance leg only in the folded configuration $q_2 = \pi$, where the swing foot coincides with the stance foot and moves perpendicular to the legs. On a flat tread the foot therefore dips into the step unless the stance leg is vertical at that instant, $q_1 = \pi/2$, which happens at $\theta_f = \frac{1}{2} + \alpha/\beta$ and requires $\beta > 2|\alpha|$; the slope model of the report ($\theta_f = \frac{1}{2}$, $q_1 = \pi/2 + \alpha$) lets the foot cross the stance point along the slope line, i.e. $(b - d)\tan\alpha$ inside the tread behind the stance foot (Fig. 10). (b) The swing foot moves on a circle of radius l around the hip, whose height is at most l (vertical stance leg), so the foot can only be above the next riser, at horizontal distance d, if $l - \sqrt{l^2 - d^2} \geq h$, i.e. $d \geq \sqrt{2h - h^2}$ in leg units, while landing on that tread needs $d \leq b$. Fig. 9 maps the admissible landing window $(b - \sqrt{2h - h^2})/b$ over stair pitch and aperture. Human stairs seen by the perception layer (rise 12 cm, run 28 cm, $\alpha = 23°$) need $\beta > 46°$, i.e. a stride of $0.79\, l$ (2.6 steps at once for a 1 m leg, or a 0.38 m leg for one step at a time), and even then a window of a few centimetres. The report's proof-of-concept staircase ($b = 1$, $h = 0.5$, $d = 0.9$) sits exactly at the boundary: window $0.134\, b$, $\theta_f = 0.89$.

3 Experiments

All numbers are produced by `python3 experiments/run_experiments.py` and `fig_perception.py` and stored in `experiments/out/metrics.json`, `perception_results.json`, `gaits.json`. Seeds are fixed; there is no randomness in the controller pipeline.

3.1 Perception

Table 1: Stair geometry from the five archived scenes versus the tape measure (mean of the two measured steps). MAE over the five scenes; the poster (iterative RANSAC, its own five test cases) reported 2.91 cm / 5.1 cm.

scene	tape h cm	D435 h cm	mono h cm	tape b cm	D435 b cm	mono b cm	pitch deg
WE 17:51:34	14.2	16.5	10.7	27.1	19.3	25.9	47
WE 18:27:47	19.5	19.6	19.9	25.3	25.1	23.3	67
WI 17:41:57	12.0	15.1	15.5	29.6	31.2	33.5	56
WI 17:47:26	12.0	14.2	12.0	29.6	28.1	28.3	53
WI 18:12:19	15.0	15.7	18.0	32.1	33.3	27.7	53
MAE		1.7	2.1		2.5	2.5	
accuracy		86.7 %	84.9 %		91.2 %	91.4 %	

Table 1 and Fig. 1 summarise the result. Rise is recovered to 1.7 cm MAE from the RealSense cloud and 2.1 cm from the monocular cloud; run to 2.5 cm from both. The largest errors are the WE 17:51:34 scene (camera pitched only 47°, risers seen at grazing angle: the riser lattice locks onto a 19 cm harmonic on the D435 cloud while the tread-edge estimate gives 25.6 cm) and the WI 17:41:57 scene, where both clouds return a 3 cm larger rise than the tape (the treads there are worn concrete with a raised nosing strip). The monocular cloud is metrically consistent with the RealSense cloud on all scenes (pitch within 8°, offsets within 3 cm), which is the point of the poster.

3.2 Flat ground: reproduction and verification of the report

With $\beta = 0.316637$, $v_1 = -0.894373$, $v_2 = 1.9$ and the reflection impact map, the linear design returns $a = [2.8250, 0.4246, -7.3543, 30.1847, -35.9442, 13.3225]$ (ascending powers), identical to the report to 4.4×10^{-5}, and $\mathcal{I} = \begin{bmatrix} -0.6864 & 0.3621 \\ -0.1459 & -0.1081 \end{bmatrix}$, $\delta = 0.3364$, $M^- = 0.1214$, $V^- = -0.2543$, $V_{\max} = 3.420$, $\min_\theta |B^\perp D\sigma'| = 0.100$, hybrid-invariance residual 2×10^{-14}, $\dot\theta^{+*} = 2.652$, $\dot\theta^{-*} = 7.883$ (Fig. 3–Fig. 4). Simulated from the limit cycle the walker takes 25 steps with period 1.044 s, stride 0.3153 m, speed 0.302 m/s, hip torque 110 N m peak / 13.5 N m rms, $|y| < 3 \times 10^{-11}$, 8.9 J of actuator work per step, and $\dot\theta^-$ constant to 2×10^{-3}. Started at 1.2× and $1.6 \times \dot\theta^{+*}$ the measured return-map slope is 0.9329 and 0.9327 against the analytic $\delta^2/M^- = 0.9326$ (Fig. 5); this slope is the Floquet multiplier, so a velocity error decays by only 7% per step and 25 steps remove 83% of it, consistent with the slow convergence visible in the report's phase portraits. The energy margin at the fixed point is only 0.09 above $V_{\max}$: started at $0.8 \times \dot\theta^{+*}$ the walker stalls before the apex, and the on-manifold sweep (Fig. 6) shows the basin is one-sided, $\dot\theta_0/\dot\theta^{+*} \in [1.0, 3.0]$ (25 steps completed, upper end of the sweep). Off the manifold (both joints offset by ϵ at $1.2 \times \dot\theta^{+*}$) the walker survives $\epsilon \in [-0.26, 0]$ rad and fails for any positive offset. The report's statements on disturbances (Section F) only tested faster initial conditions, which is the tolerant side.

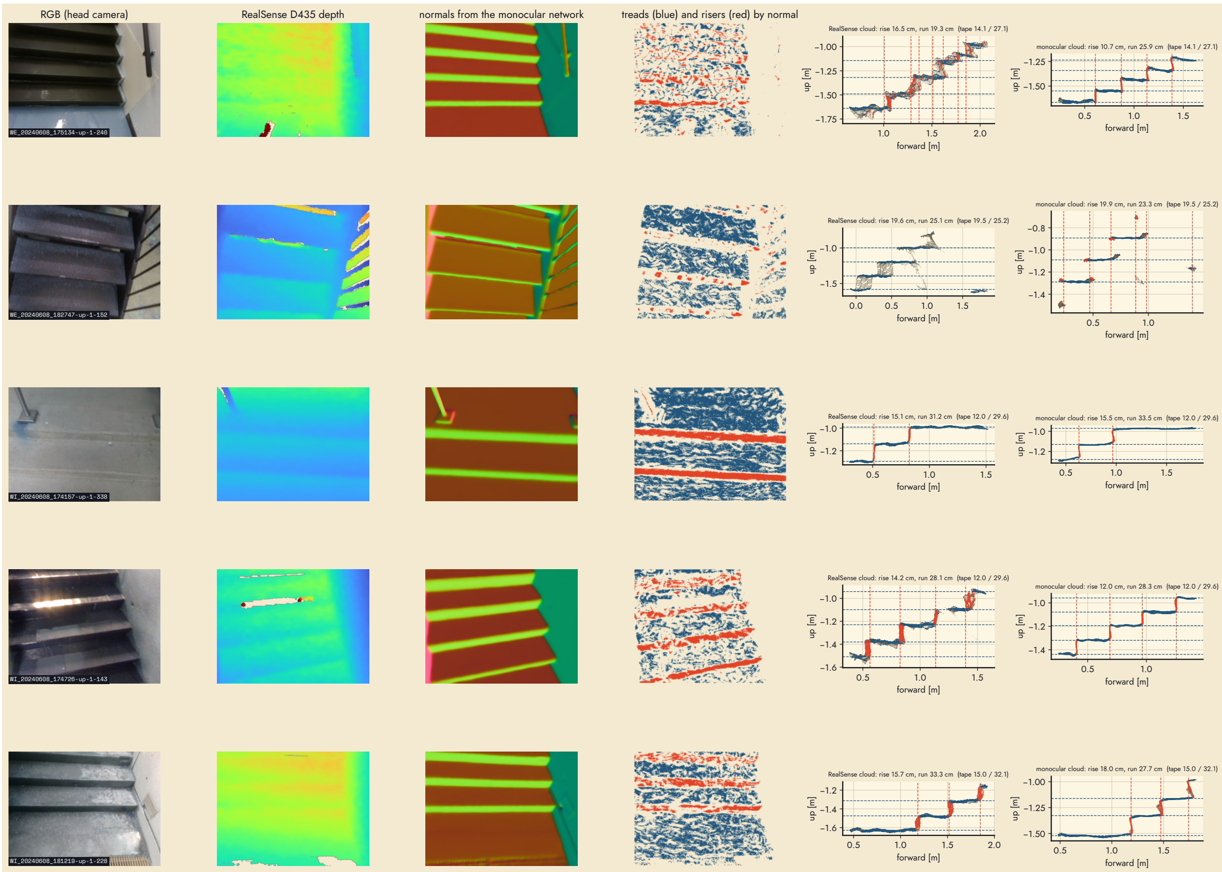

Figure 1: Per scene: RGB, RealSense depth, monocular normals, tread/riser labels from the normals, and the side profile of a 0.5 m band of the cloud with the fitted tread (blue) and riser (red) lattices, for the RealSense and the monocular cloud.

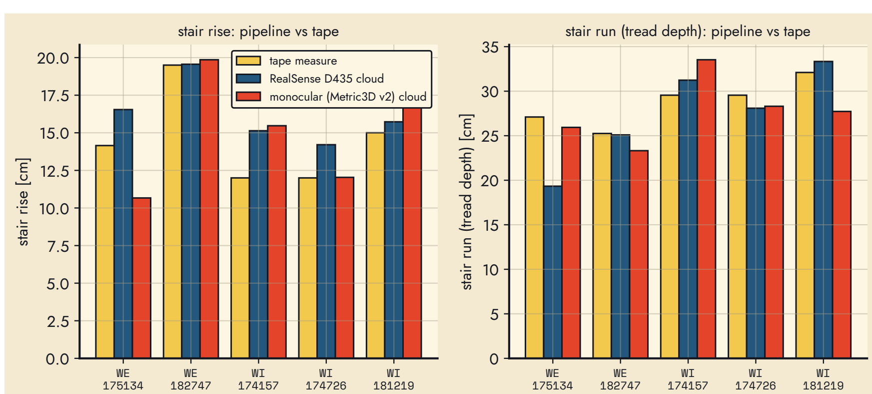

Figure 2: Rise and run per scene: tape, RealSense cloud, monocular cloud.

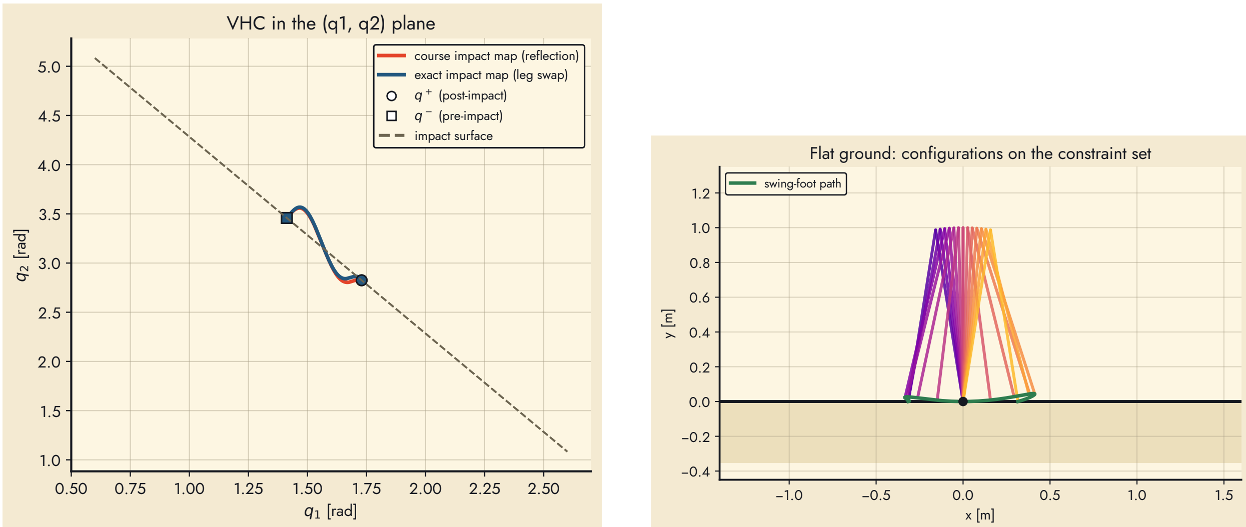

Figure 3: Left: the VHC in the (q_1, q_2) plane for the report's parameters under the reflection impact map (red) and the exact leg swap (blue); the exact map moves $\phi'(0)$ from 0.42 to 1.25. Right: the constraint set as a strobe with the swing-foot path.

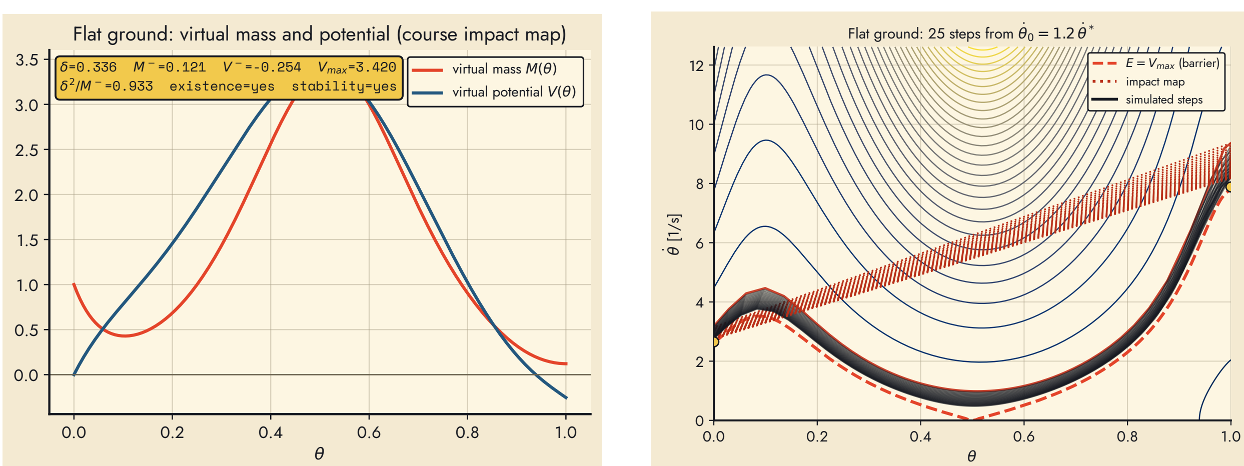

Figure 4: Left: virtual mass and potential of the report design. Right: energy levels of the constrained dynamics with 25 simulated steps from $1.2\,\dot\theta^{+*}$ (first step red, impact map dotted, fixed point yellow).

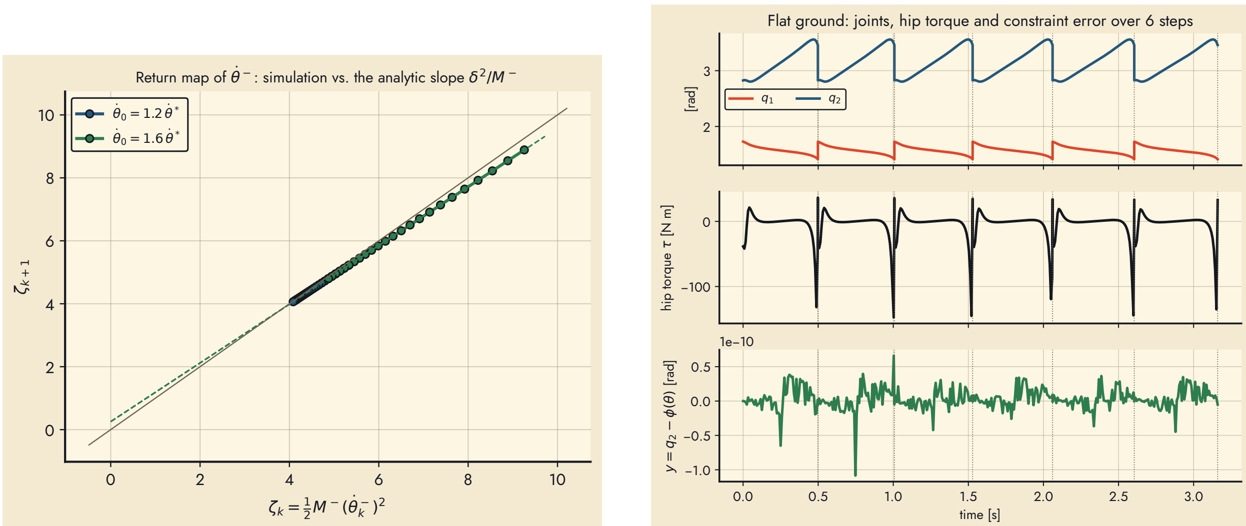

Figure 5: Left: measured return map of ζ_k for two initial speeds against the analytic affine map. Right: joint angles, hip torque and constraint error over six steps.

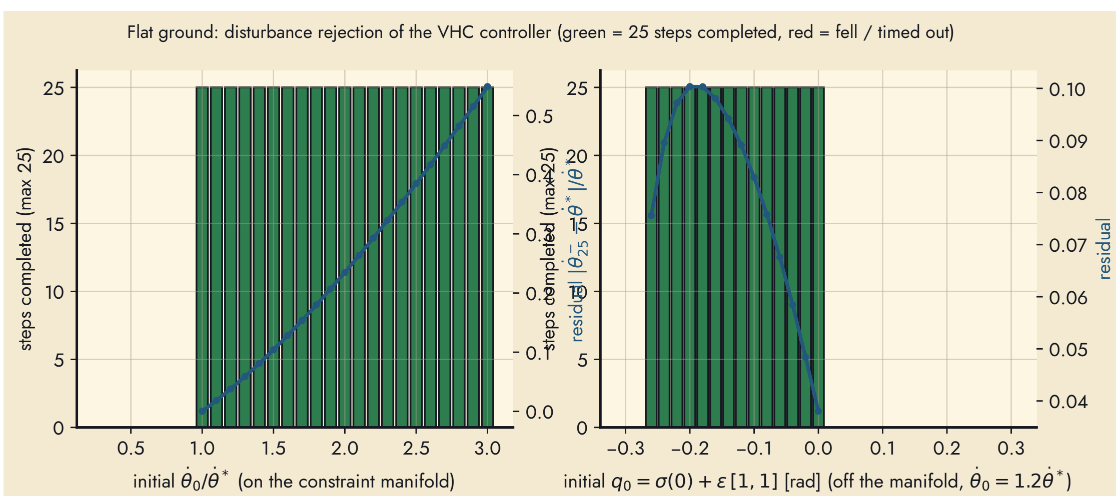

Figure 6: Disturbance sweeps on flat ground (25-step budget). Left: initial phase velocity on the manifold. Right: joint offset off the manifold.

3.3 Impact-map convention

Table 2 compares the two relabellings for the report's parameters. Under the exact leg swap the new stance leg keeps $-0.036\,\dot{\theta}^-$ instead of the reflection's $-0.107\,\dot{\theta}^-$ (δ drops from 0.336 to 0.115), and the gait fails the energy test by a wide margin: no $k = 6$ design is feasible for $\beta \in \{0.2, 0.32, 0.45, 0.6, \pi/4\}$ over $v_1 \in [-1.6, 2\beta]$, $v_2 \in [2\beta, 6]$ (45×45 grid). Retention improves with swing-leg retraction ($v_1 > 0$, Fig. 7) and the optimiser finds exact-map flat gaits with $k = 8$ at $v_1 \to 2\tilde{q}_1$ (the foot touches down with zero vertical velocity), $\delta^2/M^- \approx 0.90$; they are marginal and were not used further. On the $-10°$ slope, however, the exact map yields a comfortable retraction gait ($v_1 = 1.36$, multiplier 0.44, hip torque 12 N m peak against 204 N m for the reflection gait, Table 3).

Table 2: Report parameters ($\beta = 0.3166$, $v_1 = -0.8944$, $v_2 = 1.9$) under the two velocity relabellings.

relabelling	$\mathcal{I}_{11}$	$\mathcal{I}_{12}$	δ	M^-	V^-	$V_{\max}$	δ^2/M^-	stable orbit
reflection $R = -I$ (course)	-0.686	0.362	0.336	0.121	-0.254	3.42	0.933	yes
exact leg swap R_{kin}	0.832	-0.254	0.115	0.061	-0.203	1.67	0.214	no

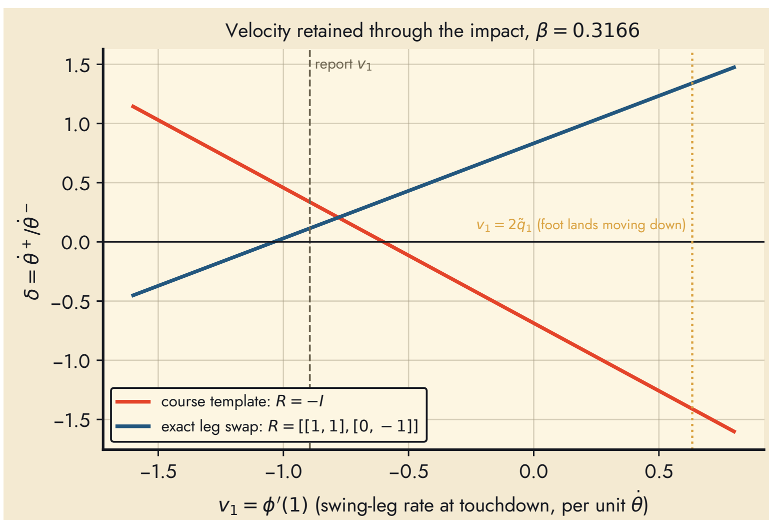

Figure 7: Phase-velocity retention δ through the impact as a function of the swing-leg rate at touchdown, for both conventions.

3.4 Slopes

Gaits were optimised for $\beta = \pi/4$ as in report Part 3 with $\alpha \in \{-10, -5, +5, +10\}°$ from seven starts, $k \in \{6, 8\}$, and accepted only if they walk 25 steps in the collision-checked simulation (Table 3, Fig. 8). Downhill gaits are found for both conventions; uphill only $+5°$ under the reflection ($k = 8$, multiplier 0.95, i.e. barely stable), and none at $+10°$: the virtual potential V^- that the impact has to balance becomes positive as the slope rises, and the polynomial family cannot make it negative while clearing the ground. The report's $+10°$ and $+30°$ cases could not be reproduced with the stability test enforced; the report shows their VHC curves and strobes but no return map.

Table 3: Slope gaits ($\beta = \pi/4$, stride 0.765 m), 12 simulated steps from $1.15\,\dot{\theta}^{+*}$.

case	map	k	v_1	v_2	δ^2/M^-	period [s]	speed [m/s]	τ_{peak} [N m]	steps
-10°	refl.	6	-2.02	1.72	0.380	1.03	0.75	204	12/12
-10°	exact	6	$+1.36$	1.76	0.443	1.00	0.78	12	12/12
-5°	refl.	6	-2.55	1.74	0.575	0.93	0.85	242	12/12
$+5^\circ$	refl.	8	-3.32	3.29	0.950	0.59	1.29	90	12/12
$+10^\circ$	both	–	no feasible design (14 starts per convention)						

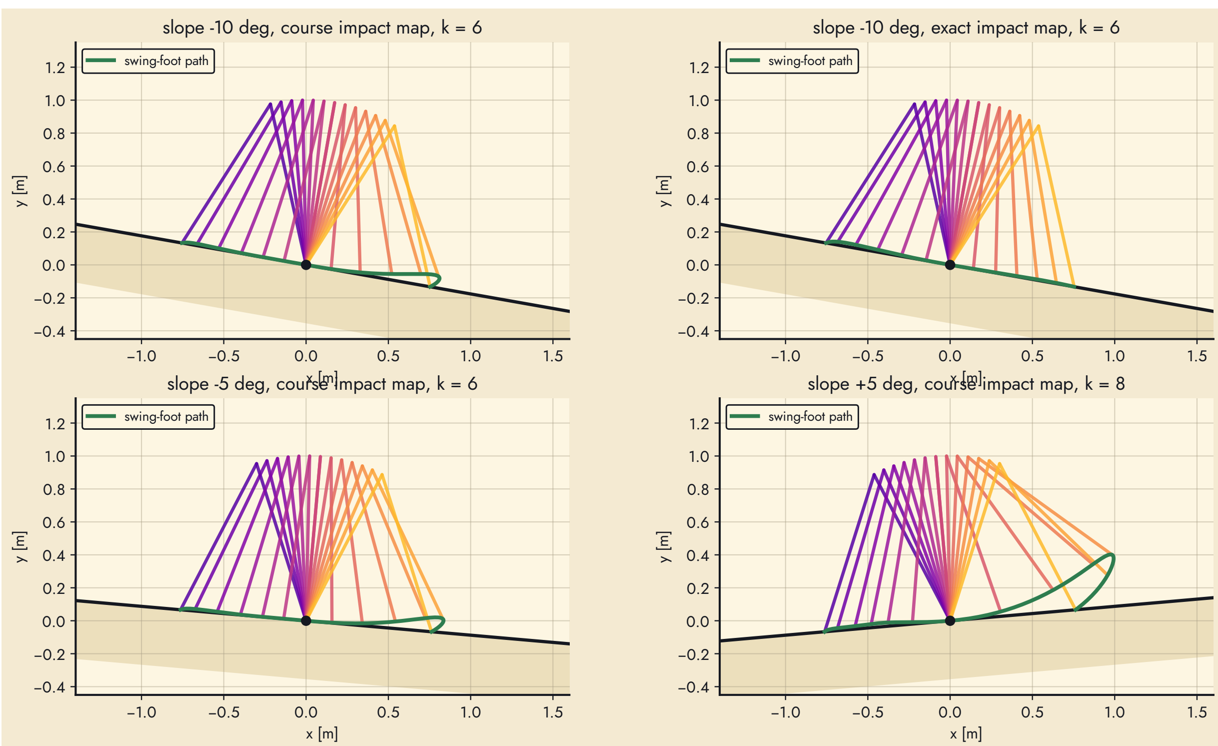

Figure 8: Constraint sets of the four slope gaits with their swing-foot paths.

3.5 Stairs

Geometry. Fig. 9 is the map of Section 2.6; Fig. 10 shows a slope-model curve for the report's staircase against the real treads: the swing foot is 4.8 cm inside the step at $\theta = 0.46$ (prediction $(b-d)\tan\alpha = 5.0\,\text{cm}$).

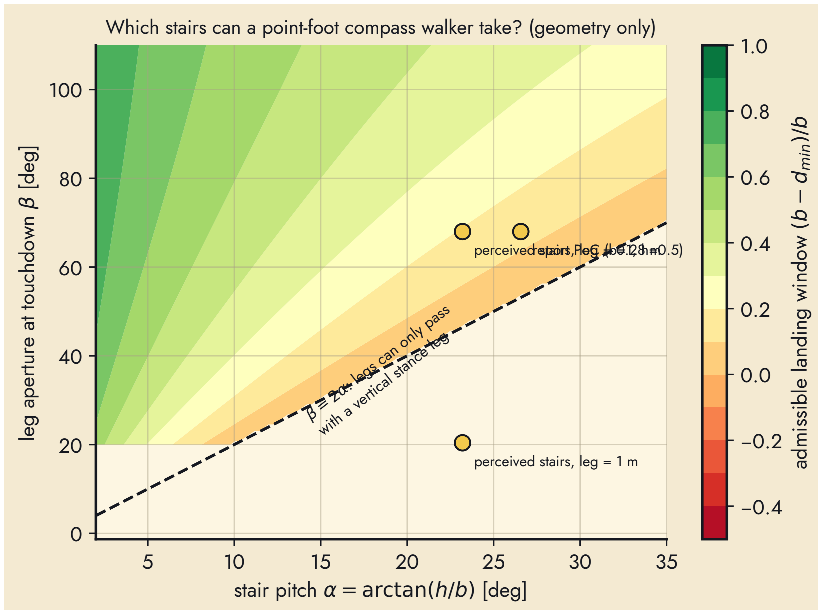

Figure 9: Admissible landing window $(b-d_{\min})/b$ for a point-foot compass walker versus stair pitch and touchdown aperture; white where the legs cannot pass with a vertical stance leg ($\beta \le 2\alpha$).

Optimised stair gaits. With the vertical-fold condition $\theta_f = \frac{1}{2} + \alpha/\beta$ and full tread/riser clearance, 80 cells $\{5, 10, 15, 20, 26.6\}^\circ \times \{45, 60, 68, 80\}^\circ \times \{\text{up}, \text{down}\} \times \{\text{reflection}, \text{exact}\}$ with $d = 0.8b$ and 10 starts each (`sweep_stairs.py`) produced no feasible design, nor did the batch for the report's staircase and for the perceived staircase scaled to a 0.28 m leg (`design_all.py`). The binding constraints are the vertical-fold condition together with the riser clearance, which force the swing leg to travel almost the whole stride while the stance leg stays within a few degrees of vertical; the resulting ϕ' breaks transversality or the energy test.

Slope gaits on real staircases. The slope gaits of Table 3 were then run on staircases of the same pitch and stride (rise 6.7 cm, run 76 cm at 5°; rise 13.3 cm, run 75 cm at 10°) while sweeping the foot placement d/b (Fig. 11, Table 4). Ascending works iff $d/b \ge 0.95$: landing within 5 % of the tread depth from the nosing puts the rear nosing so close behind the stance foot that the fold's dip stays within the 1 mm scuff tolerance, and the leading foot clears the next riser because it arrives from above with retraction. Descending works iff $d/b \le 0.05$: standing on the nosing, the swing foot leaving the fold downward is already past the tread. Both rules are direct consequences of Section 2.6. In the accepted placements the walker takes 12/12 steps with zero collisions; ascending at 1.02 m/s (climb rate 0.089 m/s, 77 N m peak), descending at 0.71 m/s (189 N m reflection, 9.3 N m exact map).

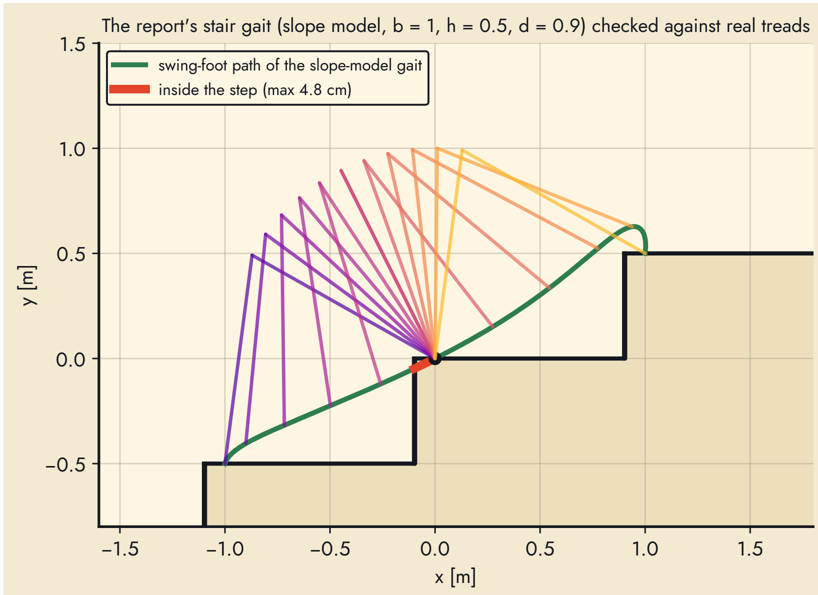

Figure 10: The slope model's swing-foot path on the report's staircase ($b = 1$, $h = 0.5$, $d = 0.9$): the highlighted part of the path is inside the step behind the stance foot.

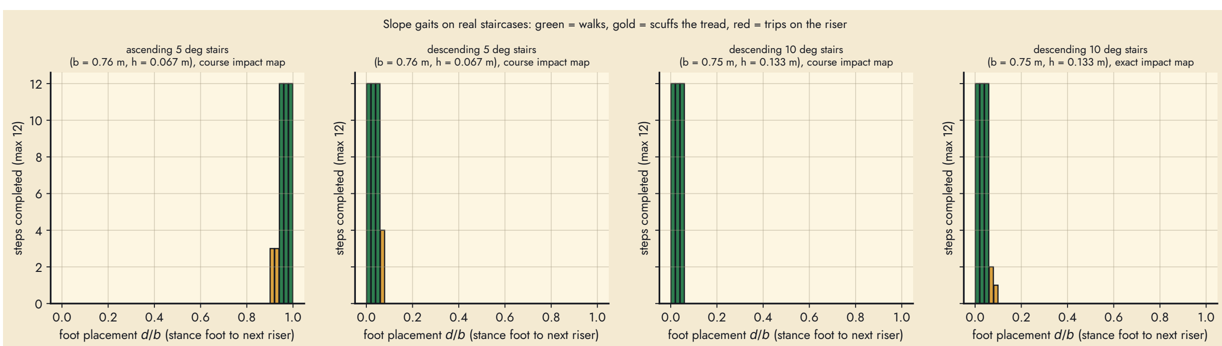

Figure 11: Slope gaits on real staircases: steps completed (of 12) versus foot placement d/b.

Table 4: Slope gaits on collision-checked staircases at the chosen foot placement.

stairs	b / h [m]	gait (map)	d/b ok	used	period [s]	speed [m/s]	τ_{peak} [N m]	steps
up 5°	0.76 / 0.067	+5° (refl., k=8)	$[0.95, 0.99]$	0.97	0.75	1.02	77	12/12
down 5°	0.76 / 0.067	−5° (refl.)	$[0.01, 0.05]$	0.03	0.99	0.77	217	12/12
down 10°	0.75 / 0.133	−10° (refl.)	$[0.01, 0.05]$	0.03	1.07	0.71	189	12/12
down 10°	0.75 / 0.133	−10° (exact)	$[0.01, 0.05]$	0.03	1.04	0.73	9.3	12/12

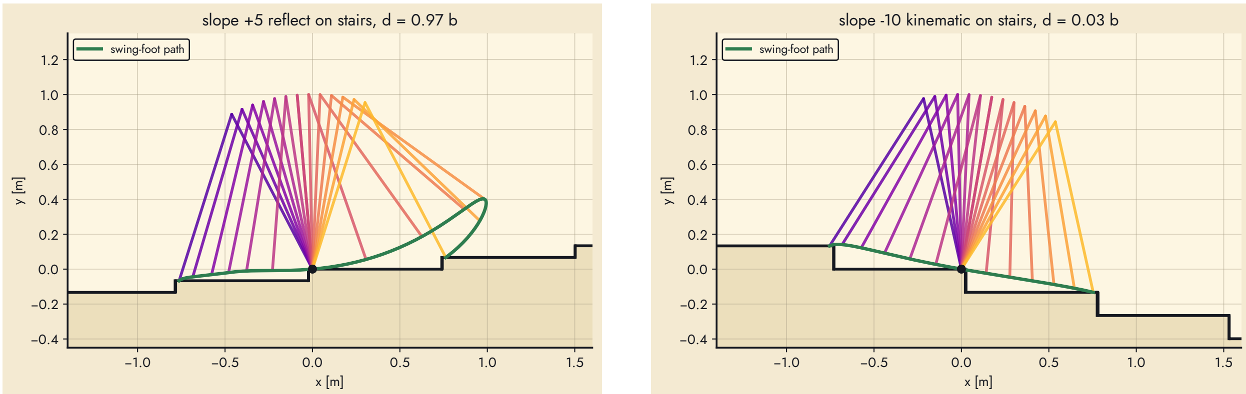

Figure 12: Left: ascending the 5° staircase with $d = 0.97b$. Right: descending the 10° staircase with the exact-map gait and $d = 0.03b$.

4 Five-link biped: knees, torso, and stepping over obstacles

The two-link acrobot of Sections 2.2–2.6 cannot lift a foot. The human-like walker of the project is the five-link biped of ECE1658 Assignment 1 (`control/MATLAB/1/biped_modeling`), which has the RABBIT structure: two legs with knees, a torso, point masses at the joints, point feet, four actuators (both knees, the hip, the torso) and an unactuated stance-foot pivot. Its parameters are those of the assignment (Table 5).

Table 5: Five-link biped parameters (ECE1658 Assignment 1, Table 1) and the coordinates used throughout.

l_1 shank	0.5 m	q_1	absolute angle of the stance shank (foot $\to$ knee), CCW from $+x$
l_2 thigh	0.5 m	q_2	stance knee, relative (shank $\to$ thigh); $q_2 > 0$ is flexion
l_3 torso	0.3 m	q_3	hip, relative (stance thigh $\to$ swing thigh)
$m_1 = m_5$ feet	1.32 kg	q_4	swing knee, relative (thigh $\to$ shank); $q_4 < 0$ is flexion
$m_2 = m_4$ knees	13.2 kg	q_5	torso, relative to the stance thigh
m_3 hip, m_6 torso mass	7.9 kg, 13.2 kg	B	$[0_{1\times 4};\ I_4]$, $B^{\perp} = [1\,0\,0\,0\,0]$

The masses are the assignment's point masses (0.05, 0.5, 0.3, 0.5 kg) scaled by 50/1.9 to a 50 kg robot; the zero dynamics and every gait design are invariant to this scaling, the torques scale with it (the regular gait peaks at 53 N m). The four actuators are rated at 400 N m continuous, the bound every periodic gait is designed to, and 800 N m peak for transients shorter than 0.1 s (the controller and the MuJoCo motors saturate there).

4.1 Model

With point masses only, everything is closed form from the segment chains: $r_2 = l_1 u(\phi_1)$, $r_3 = r_2 + l_2 u(\phi_2)$, $r_4 = r_3 + l_2 u(\phi_3)$, $r_5 = r_4 + l_1 u(\phi_4)$, $r_6 = r_3 + l_3 u(\phi_5)$ with $u(a) = (\cos a, \sin a)$ and cumulative angles $\phi_1 = q_1$, $\phi_2 = q_1 + q_2$, $\phi_3 = \phi_2 + q_3$, $\phi_4 = \phi_3 + q_4$, $\phi_5 = \phi_2 + q_5$. Then $D = \sum_i m_i J_i^\top J_i$, $C\dot q = \sum_i m_i J_i^\top (\dot J_i \dot q)$ with the bias acceleration $\dot J_i \dot q = -\sum_{\text{seg}} l\, \dot\phi^2 u(\phi)$, $G = g \sum_i m_i J_{i,y}^\top$, and the unpinned $\bar D$ and $E = [\partial r_5/\partial q \quad I_2]$ enter the same impact map (3). These closed forms agree with the sympy derivation of the assignment to 3.6×10^{-15} (`Model5.check_closed_forms`), the impact conserves angular momentum about the new stance foot to 1.3×10^{-15}, and the relabelling

$$\tilde{q} = Rq + d, \qquad R = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 \end{bmatrix}, \quad d = \begin{bmatrix} -\pi \\ 0 \\ 2\pi \\ 0 \\ -\pi \end{bmatrix}, \tag{6}$$

was verified to reproduce every physical point after the swap (1.2×10^{-15}). The archived `impact_map.m` has $R_{25} = 1$ (a $+q_5$ term in the new stance-knee angle), which that check rejects; the assignment's own Fig. 2 gives $\tilde{q}_2 = -q_4$.

4.2 Gait design

The phase of a step is the angle of the *virtual stance leg* (the line from the stance foot to the hip), $\vartheta(q) = q_1 + \epsilon(q_2)$ with $\epsilon(q_2) = \operatorname{atan2}(l_2 \sin q_2,\, l_1 + l_2 \cos q_2)$, normalised to $\theta = (\vartheta^+ - \vartheta(q))/\Delta$, $\Delta = \vartheta^+ - \vartheta^-$. Unlike the stance-shank angle used for the acrobot, ϑ keeps sweeping monotonically while the stance knee flexes and extends, so the regularity term $B^\perp D\sigma'$ (the angular momentum about the stance foot per unit phase rate) stays away from zero for knee-heavy steps such as stair climbing and stepping over a box[1]. The constraint is $\sigma(\theta) = [\,\vartheta^+ - \theta\Delta - \epsilon(\phi_2(\theta));\ \phi_2(\theta); \ldots; \phi_5(\theta)\,]$ with each ϕ_k a Bézier polynomial of degree N ($N = 6$) with coefficients $\alpha_{k,0..N}$. Given a touchdown posture q^- (Section 4.2) the post-impact posture is $q^+ = Rq^- + d$; the end coefficients enforce $\phi_k(0) = q_k^+$, $\phi_k(1) = q_k^-$, $\phi_k'(1) = v_k$ and hybrid invariance $\sigma'(0) = \lambda\, w$ with $w = \mathcal{I}\sigma'(1)$ and $\lambda = -\Delta/(w_1 + \epsilon'(q_2^+)w_2)$, so that $\delta = 1/\lambda$; the $N-3$ interior coefficients per joint, the four touchdown tangents v_k and the three posture parameters are the design variables. The zero dynamics, M, V, δ, the affine return map and the controller are the same formulas with $B^\perp = [1\,0\,0\,0\,0]$, $H = \partial y/\partial q = [\,0\ I_4\,] - \phi'(\theta)\,\partial\theta/\partial q$ and four torques. Two closed-form by-products of the analysis are used as design constraints: the feed-forward torque along the zero dynamics, $B\tau = D(\sigma'\ddot{\theta} + \sigma''\dot{\theta}^2) + C\sigma'\dot{\theta} + G$ with $\dot{\theta}^2 = (\dot{\theta}^{+2} - 2V)/M$, and the step period $T = \int_0^1 \sqrt{M/(\dot{\theta}^{+2} - 2V)}\, d\theta$.

Touchdown posture. q^- is parametrised by the stance-knee flexion $\kappa_s = q_2^-$, the swing-knee flexion $\kappa_w = q_4^- \le 0$ and the torso lean γ; the two "virtual legs" (foot $\to$ hip, hip $\to$ foot) then have lengths $L = \sqrt{l_1^2 + l_2^2 + 2l_1 l_2 \cos\kappa}$ and the swing foot is placed on the stride vector (b, h) of the terrain by the two-link inverse kinematics of the triangle (stance foot, hip, swing foot). This makes stairs and boxes ordinary strides.

Constraints. Transversality ($\min_\theta \operatorname{sgn} \cdot B^\perp D\sigma' \ge \sqrt{\zeta}$, no sign change), existence $\delta^2/M^- \le 0.95$, $V^- < 0$, a 5 % energy margin over $V_{\max}$, no knee hyper-extension ($q_2 \ge 0$, $q_4 \le 0$ along the step), torso within 35° of vertical, hip above $0.66\,l$, swing-foot lift below 0.35 m, the swing thigh below horizontal, forward progress of the foot, and clearance of the swing foot over the terrain (treads, risers, the box faces) except at the two contacts. Objective: $\int \|\sigma''\|^2$ plus the foot path length. Terrain gaits are warm-started from the flat gait of the same stride; the analysis clips the transversality denominator away from zero so that a step outside the feasible set still returns finite numbers to the optimiser.

[1] With the shank angle as phase, the trailing step of a step-over lost transversality in every design attempt: the shank hardly rotates while the knee does the work, so $B^\perp D\sigma'$ crossed zero. Switching the phase variable removed the problem without changing anything else in the framework.

Chaining gaits through the impact. A plan is a sequence of different steps. Each step's constraint is re-anchored to the actual previous impact: its q^+ is the relabelled touchdown posture of the previous gait and its initial tangent is $\lambda\mathcal{I}_{\text{prev}}\sigma'_{\text{prev}}(1)$, so the whole chain is hybrid invariant; the interior coefficients, end posture and tangents come from a periodic gait designed for that step's terrain (`build_gait(..., chain_from=...)`). The incoming phase velocity follows from the previous step's fixed point through energy conservation on the zero dynamics, $\frac{1}{2}M^-\dot\theta^{-2} = \frac{1}{2}\dot\theta^{+2} - V^-$, and the step is accepted if its barrier $V_{\max}$ is below the incoming energy.

Planner (perceive → plan → walk). From the current pose the perception layer gives the first obstacle's front distance x_o, depth d_o and height h_o. The planner accepts $h_o \le 0.25\,\text{m}$, $d_o \le 0.30\,\text{m}$, takes off $a = 0.25\,\text{m}$ before the box and lands $b = 0.20\,\text{m}$ past it (leading stride $S_1 = a + d_o + b$), splits the approach $x_o - a$ into $n = \text{round}((x_o - a)/0.40\,\text{m})$ equal steps in $[0.28, 0.55]\,\text{m}$ and designs that regular gait, then the leading step-over (box at a), then the trailing step-over (box now $d_o + b$ behind the stance foot) which is forced to end in the nominal regular gait, then resumes it. All gaits are VHCs; the transitions are chained as above and the plan is simulated against the *true* box, so perception error is part of the test.

4.3 Experiments

Flat ground. The nominal gait (stride 0.40 m) has $\delta = 1.203$, $M^- = 1.832$, $V^- = -0.569$, $V_{\max} = 2.037$, Floquet multiplier $\delta^2/M^- = 0.790$, transversality margin $\min|B^\perp D\sigma'| = 0.489$, touchdown posture (stance knee 0.39 rad, swing knee 0.00 rad, torso lean 0.25 rad). From its fixed point the walker takes 20 steps with period 1.235 s at 0.32 m/s, peak actuator torque 1.8 N m, rms 0.5 N m, constraint error below $2e{-}10$ rad and 0.60 J of actuator work per step. Started at 1.3× and 1.8× the limit-cycle phase velocity the measured return-map slopes are 0.7897 and 0.7897 against the analytic 0.7897 (Fig. 14); the speed basin over 16 steps is $[1.05, 3.0] \times \dot\theta^{+*}$, i.e. the gait cannot start slower than its limit cycle but tolerates a threefold overspeed.

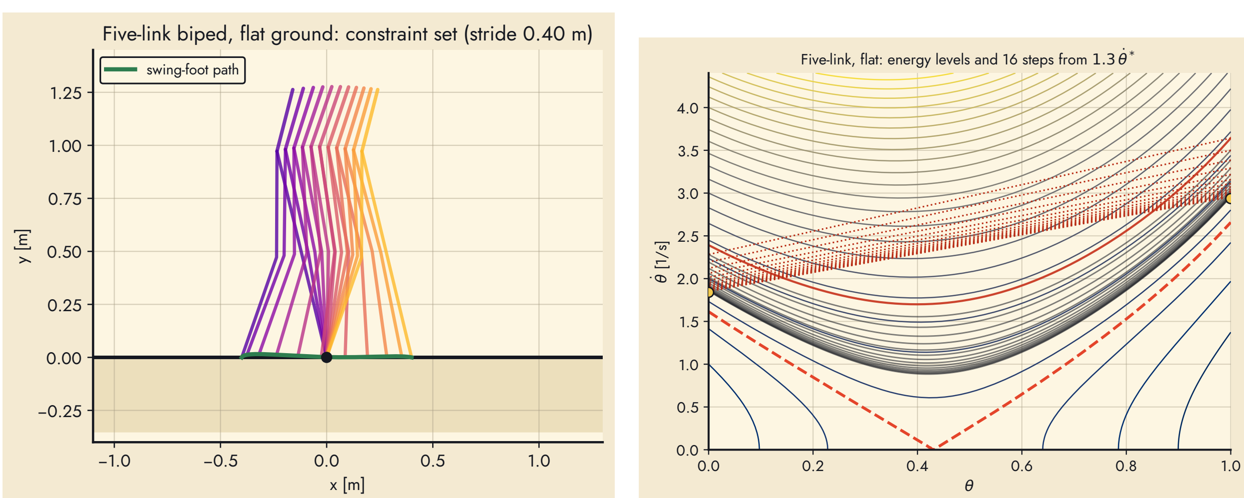

Figure 13: Five-link biped on flat ground: the constraint set with the swing-foot path (left) and the energy levels of the constrained dynamics with 16 simulated steps started 30 % fast (right).

Obstacles: perceive, plan, walk. Table 6 lists the obstacle experiments (synthetic RGB-D scenes, camera 1.3 m high pitched 35–55° down). Perception is accurate to 1 cm in distance and

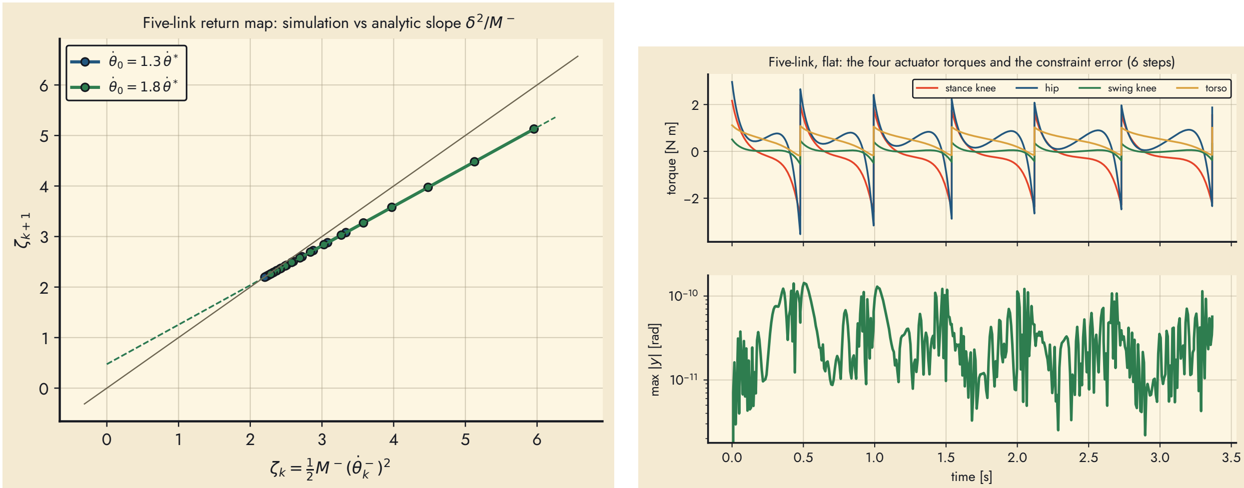

Figure 14: Left: measured return map of the five-link gait against the analytic affine map. Right: the four actuator torques and the constraint error over six steps.

height and 2 cm in depth out to 2 m; the 2.4 m box holds too few pixels. With the chaining of Section 4.4, the regular gait of Section 4.5 and the box inflated by the perception uncertainty (2 cm in front, 3 cm behind, 2 cm up), all five perceived boxes (5–22 cm high, 10–30 cm deep) are crossed end to end against the true geometry; the leading and trailing steps are designed in sequence (posture free, then ending in the regular gait; a free trailing step plus a recovery step when that is infeasible) and the take-off distance is the one the approach actually reaches. Peak torques stay at or below the 60 N m design bound and occur at the trailing crossing.

Table 6: Perceive–plan–walk experiments (true box; perceived box; plan; outcome).

case	true x_o / d_o / h_o	perceived	plan (m)	outcome
0	1.20 m / 15 / 8 cm	1.20 m / 14 / 9 cm	2×0.46, 0.63 over, resume 0.40	8/8 steps, no collision
1	1.60 m / 20 / 15 cm	1.59 m / 18 / 16 cm	3×0.44, 0.71 over, resume 0.40	5/9 steps, then trip
2	2.00 m / 25 / 22 cm	1.99 m / 23 / 23 cm	4×0.43, 0.76 over, resume 0.40	10/10 steps, no collision
3	1.00 m / 30 / 12 cm	1.00 m / 28 / 13 cm	2×0.37, 0.78 over, resume 0.40	2/8 steps, then trip
4	1.40 m / 10 / 5 cm	1.40 m / 9 / 6 cm	3×0.38, 0.54 over, resume 0.40	9/9 steps, no collision

Stairs with knees. With knees the swing foot clears the risers, so the geometric obstruction of Section 2.6 disappears; the remaining question was the energetics, since a periodic climbing gait needs $V^- < 0$ so that the impact can pay back the energy spent over the step. A first 72-start search with the stance-shank phase variable and a nearly straight landing knee found none. Two changes turned this around: the virtual-leg phase of Section 4, which keeps the constraint regular while the stance knee extends under load, and a deeply flexed landing knee as the design start. The condition $V^- < 0$ is equivalent to $\int_0^1 x_{\mathrm{com}}(\theta)\,|B^\perp D\sigma'|\,d\theta > 0$ (the virtual potential is $V = \int G_1\, B^\perp D\sigma'\,d\theta/(B^\perp D\sigma'(0))^2$), i.e. the inertia-weighted centre of mass must spend the step ahead of the stance foot: a bent landing knee puts the hip closer to the new stance foot at touchdown, and extending it while the centre of mass is ahead of the foot is exactly the knee work of human stair climbing. The resulting library (Section 4.5, Figures 17) covers runs 24–36 cm and rises 5–20 cm up and down (47 of 48 cells; the missing cell is 20 cm down on a 24 cm tread). Going up, steeper risers are met with a more flexed landing knee, a stronger torso lean and a higher lift;

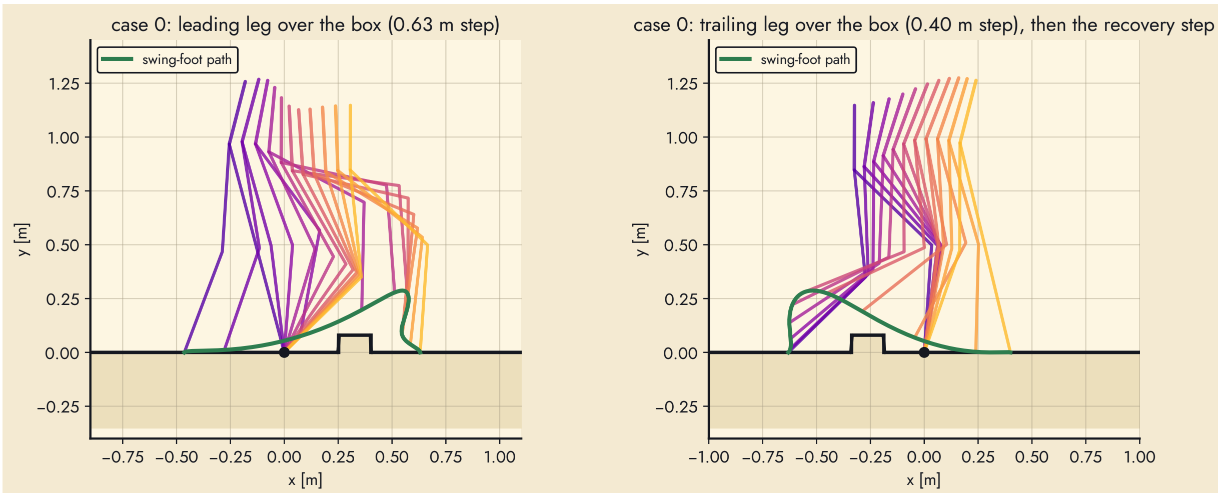

Figure 15: Case 0: the two chained step-over gaits over the perceived box (left: leading leg, right: trailing leg returning to the regular gait).

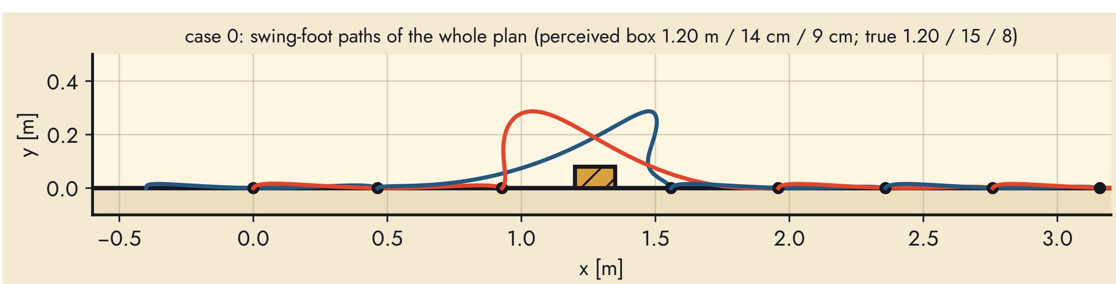

Figure 16: Case 0: swing-foot paths of the whole plan in world coordinates (alternating colours per step, dots = stance feet).

descending gaits are faster (periods 0.35–0.8 s) and land straighter. Every library cell was walked for eight steps on its uniform staircase in the collision-checked simulation. Table 7 (Section 4.7) lists the variable-staircase runs, where every stair differs from its neighbours and the planner sees noisy geometry: with the current transition machinery they stop at the first stair (the swing foot scrapes or the transition gait is not found within the iteration budget), whereas the same one-step loop in MuJoCo climbed two stairs before its third transition failed. Chaining the flat approach into the first stair remains the open problem; the periodic stair gaits themselves are not.

4.4 Chaining steps correctly

A plan is a sequence of different steps. Step k is anchored to the actual previous impact: $q_k^+ = Rq_{k-1}^- + d$ and $\sigma_k'(0) = \lambda_k \, \mathcal{I}(q_{k-1}^-) \, \sigma_{k-1}'(1)$, with λ_k from the hybrid-invariance formula of Section 4. One detail decides whether the chain is really invariant: $\sigma_{k-1}'(1)$ must be evaluated with the phase span Δ_{k-1} of the gait actually executed, which for a chained step is *not* the span of the periodic gait it was derived from. The first version of the planner used the periodic span; the post-impact velocity was then not tangent to the next constraint ($\dot{y} \neq 0$ at the switch), the tracking controller corrected it with a torque transient of 25–60 N m, and long trailing steps lost transversality. With the correct span the residual is 10^{-15} through two chained impacts and the transients disappear.

4.5 Gait library and step planner

Periodic gaits are designed offline on grids for the 50 kg robot with 400 N m actuators: flat strides (0.30, 0.40, 0.50 m), stairs on runs $\{24, 28, 32, 36\}$ cm $\times$ rises $\{5, 8, 11, 14, 17, 20\}$ cm up and down (all 24 ascending cells; 11 of the 24 descending cells — with point feet the energy of a drop can only be lost at impact, so a descending gait is a controlled fall of 0.42 s per step with an impact gain of 0.30–0.35, and the short-run cells need more than the actuators give), with the touchdown posture regularised toward the regular gait's so that the family stays smooth for interpolation, and step-overs on box heights $\{5, 10, 15, 20, 25\}$ cm $\times$ depths $\{10, 20, 30\}$ cm as leading/trailing pairs of degree-8 curves with the hand-over velocity matched to the regular gait (7 of 15 cells: every 5 and 10 cm crate and the 15 cm crate on a 30 cm footprint; higher crates exceed 400 N m in the trailing step). The design vector (three posture parameters, four touchdown tangents, the interior Bézier coefficients) varies continuously along the grid, so for an arbitrary geometry it is *interpolated* (bilinear), the gait rebuilt for the exact stride and checked; a short polish (30–60 SLSQP iterations) is run only if a constraint is violated. Online, one step is decided at a time (`step_planner.py`): the reference gait for the next foothold is looked up, chained to the executing gait, and accepted if the analytic checks pass — regularity margin, kinematics, clearance on the terrain, energy along the step for the measured incoming phase velocity (±5%), feed-forward torque; otherwise it is re-designed as a transition, first keeping the reference's landing posture (so the following step chains from a library posture), then with a free posture. A lookup costs milliseconds; a transition a few tens of seconds.

The regular gait is designed with a 30 % energy margin over its barrier (the earlier 5 % margin left the fixed point 4.8 % above $V_{\max}$, so a start only 5 % slow stalled): period 0.78 s, 0.51 m/s, multiplier 0.78, robust to starts 10 % slow.

4.6 Transitions between gait families

The steps that change family — flat to the first stair, last tread to the landing, flat onto a ramp, ramp onto the level beyond — are not periodic and cannot be interpolated from the periodic grids; improvising them online (Section 4.5) proved fragile: a transition re-designed with a free landing

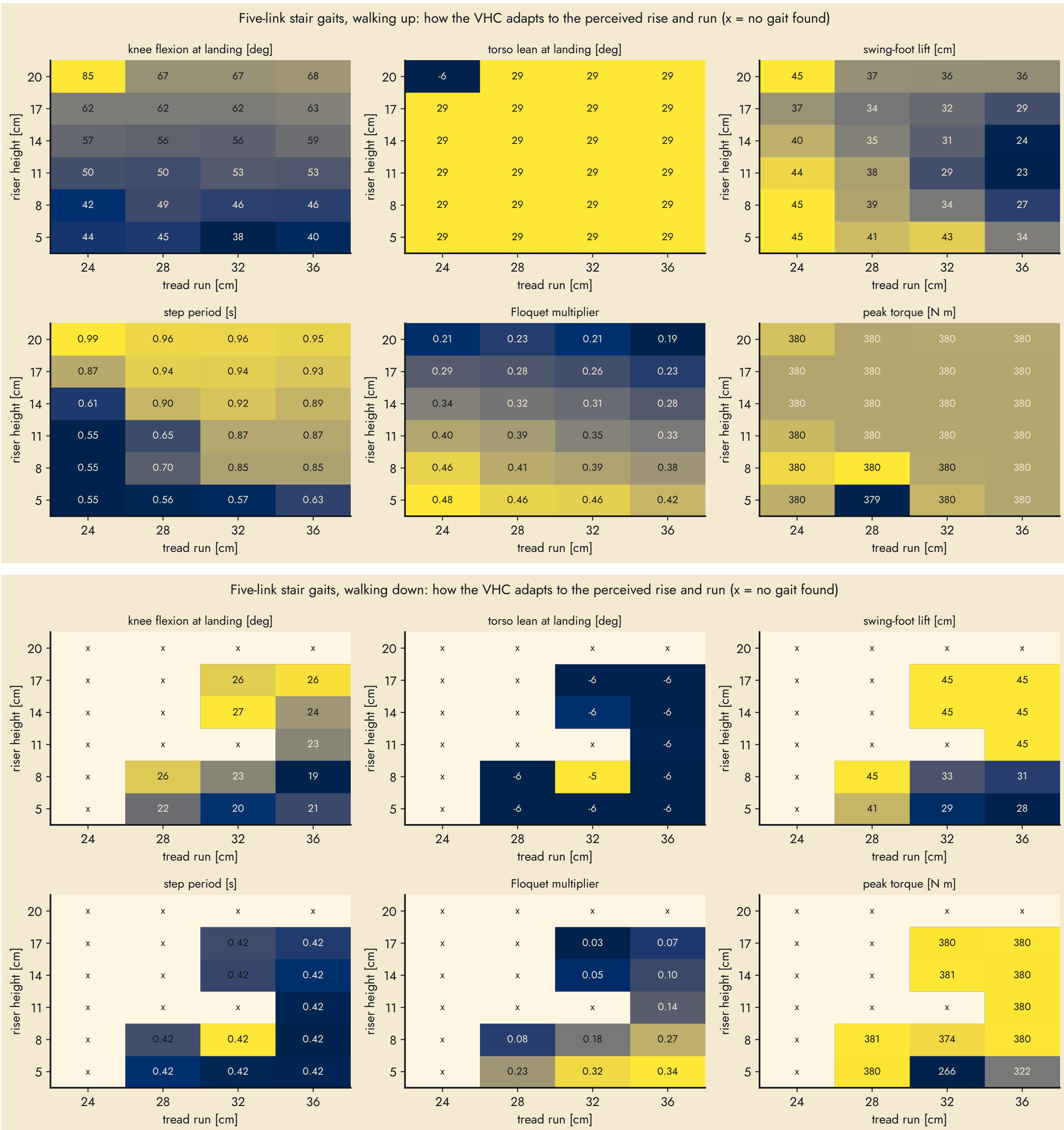

Figure 17: Stair-gait library: knee flexion at landing, torso lean, swing-foot lift, step period, Floquet multiplier and peak torque over the (run, rise) grid, walking up (top) and down (bottom).

posture leaves the family, and every periodic step after it fails its check in turn. They are therefore designed offline into a *transition library* (`transitions5_library.py`). An entry $E(b,h)$ starts where the regular gait's impact leaves the robot and lands exactly in the periodic stair gait's touchdown posture and tangents; an exit $X(b,h)$ starts where the stair gait's impact leaves it on the last tread and lands 0.40 m ahead on the landing in the regular gait's posture (the last nosing behind the stance foot is an obstacle, terrain kind `exit`); for ramps an entry per angle (kind `ramp`: flat ground until the ramp's foot, then the incline) and exits for the ramp's end 5, 15, 25 and 35 cm ahead, re-landing the regular gait at the level's height. All are degree-8 Bézier curves, robust to arrival speeds of 0.85–1.15 of nominal, bounded to 400 N m of feed-forward torque, velocity-matched to the fixed point of the gait that follows (0.90–1.12 of it; 0.45–1.15 for descending stairs, whose gaits are fast), and each cell is admitted only after the sequence regular, regular, entry, three periodic steps, exit, regular, regular, regular walks in the collision-checked hybrid simulation.

Two details of the optimisation matter more than the formulation. The warm start is the gait that follows, chained to the previous one and re-sampled at the interior nodes of the degree-8 curve; and SLSQP runs as a trust-region ladder (box radius 0.2, 0.4, 0.8 rad on the interior coefficients, then free, each stage started from the previous stage's solution). From a straight-line start, or with a free first step, the solver takes a first step that ruins an almost feasible curve (a 2 cm clearance deficit becomes a 2000 N m torque spike) on nearly every cell; with the ladder all six ramp angles and all 24 ascending stair cells verify. The entries hand over with a feed-forward spike of 500–1100 N m for 20–50 ms right before touchdown (the fixed end forces the curvature there); the actuators are therefore modelled with an 800 N m peak on top of the 400 N m continuous rating, in the analytic controller and in the MuJoCo motors alike. Two alternatives were tried and rejected: a pre-entry flat step landing in the stair family's posture (its post-impact configuration cannot match the stair gait's — the hip must sit 21 cm behind a foot that stands 11 cm higher — so the entry that follows has less than half the phase span and becomes a 0.1 s step), and a two-step entry with a free first landing (the same collapse of the phase span, now at the second step).

4.7 Closed loop in MuJoCo

The biped is modelled in MuJoCo 3.5 as a planar floating base (slide x, slide z, pitch) with two thighs, two shanks and a torso, the point masses of Table 5 placed at the joints, point feet as 12 mm spheres in frictional contact ($\mu = 1.2$, elliptic cones, no-slip iterations, contact time constant 2 ms), and four torque motors. The analytic coordinates are read from the simulator at every control step (the segment angles from the site positions, their rates from the body angular velocities), the controller of Section 4 returns the generalised torques, and they are mapped to the motors by virtual work; touchdown of the swing foot past mid-step switches the stance leg. On flat ground the simulated period converges to 0.790 s against 0.783 s analytic (with the earlier slow gait, 1.238 s against 1.236 s), the strides are exact, the tracking error is below 4×10^{-3} rad and the foot slips 0.7 cm per step; with MuJoCo's softer default contacts the period of the slow gait drifted from 1.30 to 1.6 s over twenty steps, which is why the contact settings matter. An RGB-D camera on the torso (75° vertical field of view, pitched 38° down, 320×240) is rendered every second of simulated time; its depth image is back-projected into an organized cloud and, with the camera pose from the kinematic chain, the up-facing points are clustered by height (the offset clustering of Section **??**) into floor levels whose front and back edges give riser positions, tread runs, box distance and depth, and drop edges. In the simulated scenes the estimates are within 1 cm (rise, run, riser and box distance, height) from 1 m or more away in 20–70 ms per frame; descending flights become visible only within about a metre of the edge because the treads below are occluded by it, which the world model handles by remembering the edge and waiting for the treads.

Ramps and the combined course. Ramps are walked with periodic slope gaits (`Terrain(kind='slope')`, `slope5_library.py`: $\pm 3°$, $\pm 6°$, $\pm 10°$ at a 0.40 m stride, 30 % energy margin) selected by the perceived angle; the perception fits a plane to the up-facing points that belong to no level, and a tilt above 2.5° is a ramp (its foot and top from the inlier extents; in the scenes the angle is recovered to 0.2° and the foot to within 10 cm, the first centimetre of rise being absorbed by the floor level). The planner approaches the ramp's foot with strides within 10 % of the regular one (a 0.50 m approach gait hands the entry 20–50 % more speed than it was designed for), enters with the library entry step, chains the slope gait step by step, and leaves with the library exit whose end distance is nearest to the perceived end of the ramp. The combined course (crate, four stairs up, flat, ramp down, two crates, ramp up, flat, crate, four stairs down) exercises every decision of the loop in one run; its hip trajectory and footholds are drawn in Figures 18–19.

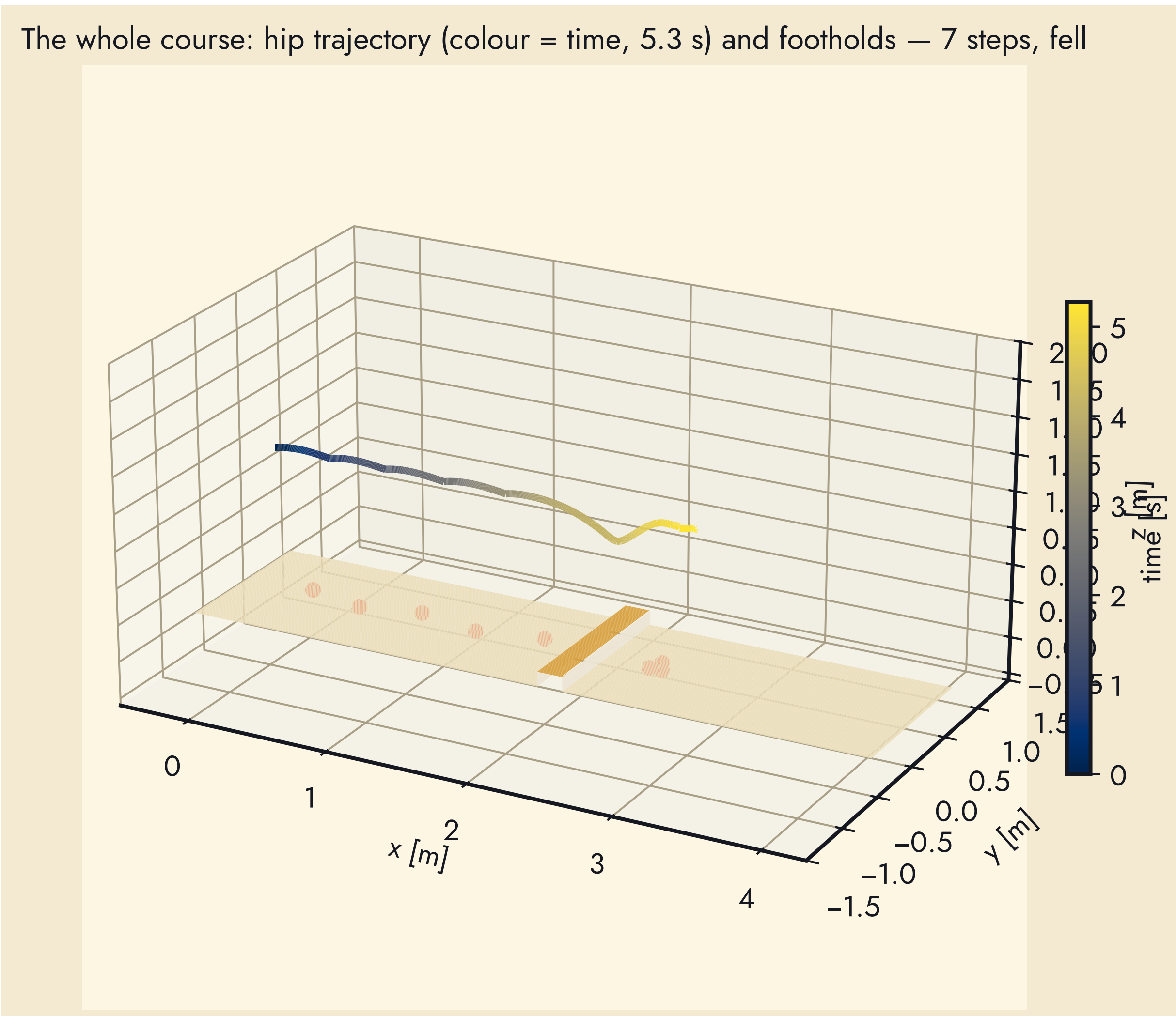

Figure 18: The combined course in MuJoCo: hip trajectory coloured by time and the footholds (red) over the crates, the flight up, the ramps and the flight down.

Staircases. Table 7 lists the variable-staircase runs.

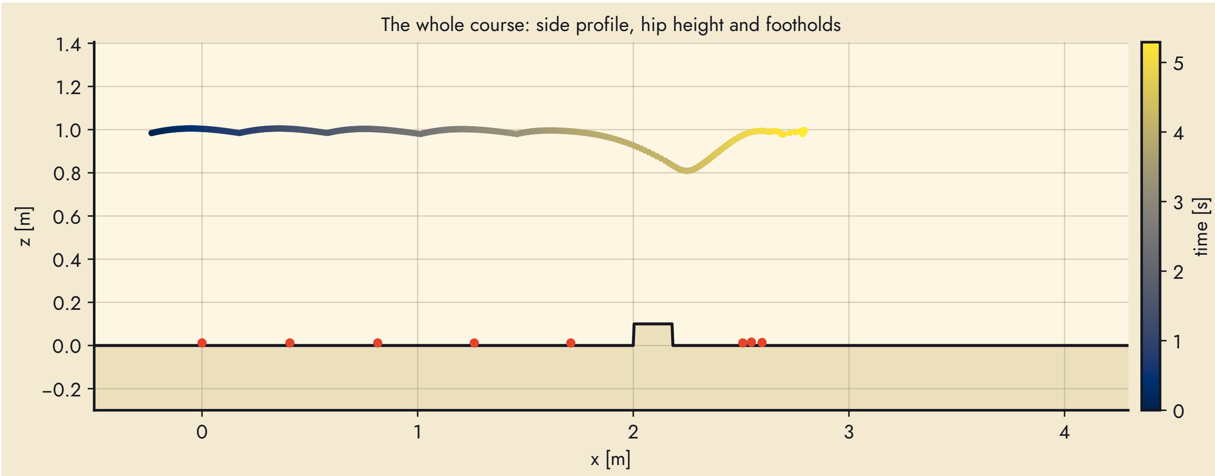

Figure 19: Side profile of the same run: ground, hip height and footholds.

Table 7: Staircases with per-stair rise and run (true geometry executed; the planner saw noisy values): gaits chained / re-designed as transitions / accepted infeasible, landing-knee range, outcome, peak torque.

run	stairs	rise [cm]	run [cm]	gaits c/r/i	knee [°]	steps	$\tau_{\max}$ [N m]
up gentle	8	6–8	32–34	7 / 0 / 1	40–47	all 14	624
up perceived	8	10–14	27–29	7 / 0 / 1	41–57	all 14	614
up steep	8	14–19	24–31	0 / 0 / 1	66–66	3/4, trip	429
up growing	10	5–19	27–33	7 / 1 / 2	44–64	all 16	800
up irregular	10	9–16	24–34	6 / 1 / 3	42–60	all 16	627
down gentle	8	6–8	33–35	2 / 0 / 1	22–23	5/6, scuff	401
down perceived	8	9–13	32–34	3 / 0 / 3	15–26	8/9, scuff	800
down steep	8	12–17	30–37	3 / 0 / 4	22–27	9/10, fell	800
down growing	10	5–16	31–37	4 / 0 / 0	19–25	6/7, scuff	539
down irregular	10	8–14	30–36	3 / 0 / 4	22–26	9/10, singular	800

MuJoCo runs. Table 8 lists the closed-loop runs.

Status. With the transition library and the acceptance rules above, the analytic closed loop completes four of the five ascending flights of Table 7 (rises of 7–19 cm, every stair different, including the flight whose rise grows from 5 to 19 cm); the fifth, with a 19 cm first stair on a 26 cm run, has no verified entry cell. The descending flights walk five to nine stairs at the library's fixed point (0.33 s per stair, about 1 m/s) and end when an interpolated cell of the sparse descending library leaves the swing foot a few millimetres over a nosing, or at the exit: no single step brakes the descent's energy (2.4 times the regular gait's phase velocity after the impact) back into the regular gait within 400 N m, and a two-step braking exit is the next thing to design. In MuJoCo the flat gait is exact, the ramp entry executes and one crate is crossed end to end; the staircase and ramp runs fall at the first family transition, where the rigid impact model over-predicts the post-impact velocity by 20–40 % and the entry's torque spike saturates the motors, and the other crates fall at the trailing step. The batch table lists every run as it happened.

5 Implementation and Usage

```
cd BipedNav/experiments
python3 stair_perception.py out/perception_results.json  # 10 clouds, ~2 s
python3 fig_perception.py                    # figures + perception_summary.json
python3 design_all.py                        # gait batch -> out/gaits.json (2 min)
python3 sweep_stairs.py                      # 80-cell stair feasibility sweep
python3 run_experiments.py                   # two-link figures + out/metrics.json
python3 design5_all.py; python3 obstacle_perception.py
python3 plan_obstacle.py                     # perceive -> plan -> walk (five-link)
python3 run_experiments5.py; python3 make_video5.py
python3 make_video.py                        # out/video/bipednav_demo.mp4
cd ../docs && make                           # this document
```

`acrobot_vhc.py` composes as `Model` (dynamics, impact) → `Gait` (design + analysis) → `Terrain` → `simulate`. Physical parameters live in `Params`; a real robot would replace `controller()`'s model terms with measured ones and `simulate()` with the hardware loop, keeping `Gait` unchanged. The MATLAB originals remain in `control/MATLAB` for reference; MATLAB itself was not required for any result (the batch launcher of R2023b returned no output on this machine). Fonts for the figures are the site's (Limelight, Bangers, Jost, Space Mono) under `experiments/fonts`.

6 Limitations

- Five scenes are too few for a statistically meaningful perception evaluation; the tape measurements were taken within minutes of the recordings but not at the exact steps in view, and the poster's own test set is not in the archive. The pipeline assumes a camera pitched down 40–70° and visible risers or tread edges.
- Both walkers are planar point-foot models with massless actuation and a plastic impact. The two-link acrobot needs 100 N m–240 N m for a 2 kg mechanism, and its flat gait (Floquet multiplier 0.93, one-sided basin) is fragile. The five-link biped is better conditioned (multiplier 0.79, about 2 N m on flat ground, 24 N m during the trailing crossing of the 8 cm box), but its

Table 8: Closed-loop MuJoCo runs: perception frames / time per frame, decisions / median / longest, outcome, peak motor torque, mean stance-foot slip per step.

scene	terrain	frames / ms	decisions / med / max	outcome	$\tau_{\max}$	slip [cm]
box	crate 12 cm high, 20 cm deep, 2.4 m ahead	6 / 45	3 / 26 ms / 0 s	6 steps, 5.4 s, fell	230	0.9
box00	crate 5 cm high, 15 cm deep, 2.0 m ahead	5 / 42	6 / 9 ms / 40 s	6 steps, 4.7 s, fell	800	3.1
box01	crate 7 cm high, 20 cm deep, 2.6 m ahead	8 / 48	3 / 12 ms / 9 s	5 steps, 7.7 s, stalled	92	0.7
box02	crate 9 cm high, 25 cm deep, 2.3 m ahead	6 / 70	4 / 16 ms / 15 s	7 steps, 5.6 s, fell	800	14.1
box03	crate 10 cm high, 12 cm deep, 3.0 m ahead	23 / 45	27 / 3 ms / 47 s	28 steps, 22.0 s	414	0.7
box04	crate 10 cm high, 30 cm deep, 2.4 m ahead	7 / 36	3 / 44 ms / 55 s	6 steps, 6.4 s, fell	304	0.8
course	crate 10 cm × 18 cm at 2.2 m, then a flight of 4 stairs (10–14 cm) at 5.0 m	6 / 20	3 / 2984 ms / 12 s	6 steps, 5.1 s, controller: torque blow-up	11	0.9
course full	crate → 4 stairs up → flat → ramp -6° → two crates → ramp +6° → flat → crate → 4 stairs down	6 / 77	5 / 19995 ms / 97 s	7 steps, 5.3 s, fell	800	0.8
down01	5 stairs down, drops 13–14 cm, runs 30–34 cm	7 / 84	7 / 8693 ms / 207 s	9 steps, 6.4 s, fell	800	5.1
down02	5 stairs down, drops 9–12 cm, runs 32–33 cm	6 / 87	7 / 13721 ms / 71 s	9 steps, 5.9 s, fell	191	0.7
down03	4 stairs down, drops 12–14 cm, runs 33–35 cm	6 / 66	4 / 125 ms / 159 s	6 steps, 5.2 s, fell	800	2.3
down04	4 stairs down, drops 7–10 cm, runs 32–33 cm	6 / 79	6 / 54090 ms / 574 s	8 steps, 5.7 s, fell	800	4.9
down05	4 stairs down, drops 11–14 cm, runs 32–36 cm	6 / 54	7 / 10136 ms / 157 s	9 steps, 5.2 s, fell	800	6.3
down06	6 stairs down, drops 5–9 cm, runs 32–34 cm	7 / 71	6 / 3994 ms / 572 s	8 steps, 6.1 s, fell	155	0.7
down07	4 stairs down, drops 10–11 cm, runs 33–35 cm	6 / 64	6 / 3488 ms / 106 s	7 steps, 5.2 s, fell	800	1.3
down09	5 stairs down, drops 6–11 cm, runs 32–35 cm	5 / 48	6 / 19472 ms / 100 s	7 steps, 4.4 s, fell	800	8.7
ramps		7 / 71	9 / 33521 ms / 166 s	10 steps, 6.7 s, fell	800	8.0
stairs down	flight of 5 stairs down, drops 10–12 cm, runs 28–31 cm, 2.6 m ahead	6 / 67	5 / 10628 ms / 228 s	7 steps, 5.5 s, fell	800	5.3
stairs up	flight of 5 stairs, rises 10–13 cm, runs 27–30 cm, 2.6 m ahead	6 / 34	4 / 23 ms / 47 s	6 steps, 5.1 s, fell	94	0.7
up00	6 stairs up, rises 15–18 cm, runs 31–33 cm	6 / 75	4 / 8 ms / 0 s	6 steps, 5.8 s, fell	92	0.7
up01	5 stairs up, rises 15–17 cm, runs 29–33 cm	6 / 87	6 / 4788 ms / 59 s	10 steps, 5.3 s, fell	800	2.5
up02	5 stairs up, rises 11–14 cm, runs 27–29 cm	6 / 70	4 / 56 ms / 96 s	7 steps, 5.6 s, fell	475	2.0
up03	5 stairs up, rises 14–17 cm, runs 29–34 cm	5 / 89	3 / 48 ms / 0 s	5 steps, 4.9 s, fell	123	0.7
up04	5 stairs up, rises 11–13 cm, runs 32–36 cm	6 / 74	4 / 4319 ms / 9 s	6 steps, 5.6 s, fell	92	0.7
up05	6 stairs up, rises 12–16 cm, runs 28–31 cm	7 / 75	5 / 70 ms / 9 s	8 steps, 6.3 s, fell	563	1.9
up06	6 stairs up, rises 13–15 cm, runs 26–31 cm	7 / 62	6 / 19 ms / 179 s	9 steps, 6.5 s, fell	800	3.3
up07	5 stairs up, rises 13–17 cm, runs 29–30 cm	7 / 68	5 / 19 ms / 8 s	8 steps, 6.3 s, fell	567	1.9
up08	4 stairs up, rises 14–16 cm, runs 29–33 cm	7 / 79	4 / 35 ms / 11 s	6 steps, 6.2 s, fell	168	0.9
up09	4 stairs up, rises 8–11 cm, runs 27–31 cm	8 / 93	7 / 20 ms / 173 s	9 steps, 7.9 s, fell	800	0.9

gaits were designed with point masses at the joints, no friction cone and no actuator limits, and the obstacle plans were executed in the same simulator they were designed for.

- The impact-convention question (Section 2.2) changes the quantitative conclusions of the report; the exact swap is derived from the kinematics and verified numerically, but the course material could define the acrobot walker differently, and this should be confirmed with the instructor before publication.
- Stair climbing at human pitch is infeasible for the two-link morphology (Section 2.6); its demonstration uses 5° and 10° staircases with 75 cm treads and a foot-placement rule that a real controller would have to enforce through step-to-step foot targeting, which is outside the VHC framework as implemented. The five-link biped clears risers geometrically, but the climbing gait itself (Section 4) is still open.
- The gait optimiser is local (SLSQP from a handful of starts) and the stair sweep is coarse; infeasibility claims are "no design found under these constraints", not proofs. For the five-link walker the binding limitations are now the transition designs (a chained step that fails the analytic checks costs 10–110 s to re-design, which is far from real time even though the perception runs at 1 Hz), the 20 cm descending cell missing from the library, and the fact that every gait was designed with point masses, no actuator model and no friction cone; the MuJoCo runs are the only test with contacts the design did not assume.
- The five-link planner chains gaits by re-anchoring periodic designs to the actual impacts without re-optimising; hybrid invariance holds but clearance and energy are only checked, not enforced, on the chained curves.
- The browser demo uses a general-purpose relative-depth network with a camera-height scale, not the fine-tuned metric network of the study; expect 10–20 % scale error on unfamiliar photos.

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